

A 2.5D APPROACH TO TRAINING ROBUST 3D POSE ESTIMATORS IN SPECIALIZED EXERCISE SETTINGS

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1 PROJECT SUMMARY

One of the biggest obstacles in developing robust, generalizable, 3D human pose estimators is the quality of annotated data. Our project seeks to investigate the relationship that diverse 2D supervised data plays when paired with scarce yet valuable 3D labeled data in facilitating more robust 3D pose estimations. Literature has shown promising results in a hybrid 2.5D approach Pavlo et al. (2019); Martinez et al. (2017) that can yield projections that are both anatomically plausible and highly generalizable.

We will evaluate the robustness of this hybrid approach by fine-tuning a pre-trained Spatio-Temporal Transformer Yang et al. (2024) on specialized fitness data with Low-Rank Adaptation (LoRA) and custom biomechanical loss functions. We hope to better understand how "in-the-wild" 2D data can complement "lab-accurate" 3D data to engineer pose estimators that perform reliably in unconstrained, high-stakes environments.

2 RELATED WORK

3D human pose estimation in video with temporal convolutions and semi-supervised training
Pavlo et al. (2019)

- This paper presents a novel approach to reconstructing 3D human pose estimations by "lifting" 2D pose estimations into 3D through a fully convolutional model architecture. This proposed architecture performs dilated temporal convolutions on the predicted 2D pose estimates to output 3D estimates. It also introduces a semi-supervised training methodology via back-projection that is used to circumvent the scarcity of 3D labeled data. This training method projects the predicted 3D pose back into 2D accurately given camera parameters and compares these back-projected points to 2D ground truth, introducing another penalty.

CameraPose: Weakly-Supervised Monocular 3D Human Pose Estimation by Leveraging In-the-wild 2D Annotations
Yang et al. (2023)

- This paper introduces a generator/discriminator pipeline to augment 2D datasets to improve 3D pose estimation. It doesn't require pose labels, and rather uses reprojection back to 2D to minimize the loss there. It introduces a camera parameter branch to the network, such that when 3D poses are estimated, the camera pose is also calculated, and a random transformation can be applied to the data, creating additional 2D data for the model. We can use this to better proportionally weight rare environments by augmenting the occurrence of those environments.

Learning 3D Human Pose Estimation from Dozens of Datasets using a Geometry-Aware Autoencoder to Bridge Between Skeleton Formats
Sárándi et al. (2023)

- This paper poses an autoencoder model that can take multiple 3D datasets and embed them in a latent space such that discrepancies in 3D dataset annotations are reconciled — e.g. we can mix multiple sources to increase the size and diversity of our dataset. The authors report that training on a combination of 3D data led to improved performance on 3D benchmarks,

especially the 3D poses in the wild benchmark, compared to prior models trained on one type of dataset.

A simple yet effective baseline for 3d human pose estimation

Martinez et al. (2017)

- This paper reports extremely competitive results using a simple architecture that can infer 3D pose coordinates from only 2D joint coordinates. The authors transform 2D imagery data into joint coordinates by effectively treating it as a 2D human pose estimation task, a fundamentally easier task to solve, and the output (and input of the 3D neural network) dataset that is both richer and lower-dimensional.

APTPose: Anatomy-aware Pre-Training for 3D Human Pose Estimation

Yang et al. (2024)

- This paper discusses APTPose, which is a framework that helps improve 3D pose estimation through Hierarchical Masked Pose Modeling and a reprojection module for integrating 2D and 3D supervision. The paper aligns with our proposed method of incorporating hybrid supervision to bridge the domain gap between lab-based datasets and the in-the-wild datasets. A notable difference, though, is that our approach uses LoRA to efficiently specialize a spatio-temporal transformer for specific occlusion patterns of gym exercises.

Our proposed method differs from prior works like the Martinez et al. baseline by incorporating temporal context, beats VideoPose3D’s convolutional approach with flexible self-attention mechanisms instead, and improves upon CameraPose and ACAE by focusing on robustness and physical constraints rather than just general-purpose accuracy.

3 PROPOSED METHOD

High-Level Strategy: Hybrid Spatio-Temporal Lifting

The proposed strategy utilizes a two-stage transfer learning and domain adaptation pipeline designed to bridge the gap between laboratory-accurate 3D labels and diverse, in-the-wild sequences. We derive this approach and portions of architecture from APTPose described in Yang et al. (2024).

The approach begins with a Spatio-Temporal Transformer backbone initialized with weights pre-trained on large-scale motion capture data (AMASS) via Masked Pose Modeling (MPM). This ensures the model possesses an inherent anatomical understanding of human motion before fine-tuning begins.

The core of the method is a 2.5D hybrid training procedure that optimizes the model using a mix of fully supervised 3D data (Human3.6M) and semi-supervised 2D gym data. To maintain computational efficiency and prevent the model from "forgetting" general human motion, we implement Low-rank adaptation (LoRA) on the attention layers. The 2D training pool consists of a 50/50 mix of gym-specific videos and general "in-the-wild" footage, where 2D keypoints are pre-extracted using a detector to serve as pseudo-labels for a back-projection consistency check.

Neural Network Specifications

The model takes a video or temporal sequence of 2D joint coordinates and outputs a sequence of 3D coordinates in a root-relative camera coordinate system. The model is optimized using a composite objective function ($\mathcal{L}$) that balances accuracy, generalizability, and physical plausibility:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{3D} + \lambda_1 \mathcal{L}_{proj} + \lambda_1 \mathcal{L}_{biomech} \quad (1)$$

1. $\mathcal{L}_{3D}$ **Supervised 3D Loss:** Mean per-joint position error (MPJPE) between predicted 3D poses and ground-truth labels from laboratory datasets.

2. $\mathcal{L}_{proj}$ **Weakly-Supervised Reprojection Loss:** Applied to unlabeled gym videos. This measures the distance between the input 2D keypoints and the 3D predictions after they are projected back into 2D space via a perspective camera model.
3. $\mathcal{L}_{biomech}$ **Biomechanical Constraint Loss:** A combination of bilateral symmetry loss to enforce equal bone lengths between left and right limbs and Anatomical Hinge Loss to penalize joint rotations that exceed physical human limits, such as knee hyperextension.

Evaluation Metrics

1. **MPJPE / P-MPJPE:** Standard millimetric error for 3D accuracy.
2. **Weakly-Supervised Reprojection Loss:** Applied to unlabeled gym videos. This measures the distance between the input 2D keypoints and the 3D predictions after they are projected back into 2D space via a perspective camera model.
3. **Bilateral Length Inconsistency (BLI):** A custom metric measuring the variance in bone lengths to quantify skeletal realism and symmetry.

Logical Reasoning: Why This Method Should Work

- By introducing 2D gym data into the training loop, the model expands its "pose understanding" to include more extreme poses that are absent in standard 3D datasets. The 3D data provides the geometric scale understanding, while the 2D gym data provides the necessary range of possible cases.
- Harnessing a spatio-temporal transformer allows us to use self-attention to correlate occluded joints in the current frame with their visible positions in previous or future frames, providing a more robust reconstruction than single-frame models, especially in a gym setting that requires a lot of interpolating for occluded frames.
- By including biomechanical losses to enforce physical constraints, we prevent the model from "cheating" during 2D reprojection. Because 2D-to-3D lifting is not a deterministic problem, standard models often stretch or distort limbs to match 2D points. These constraints force the model to produce a skeleton that is physically accurate.

4 DATASETS / ENVIRONMENTS

We want to focus on hybrid supervision for monocular 3D human pose estimation using labeled 3D and 2D datasets, all of which ensure sufficient sample size for our batched training. Training will be conducted via Georgia Tech's PACE ICE cluster for GPU resources.

- **Human3.6M:** This is a large-scale motion capture dataset including millions of frames with 3D joint annotations collected in a controlled lab environment with motion actors. We want to use this dataset as geometric ground truth when learning the structural priors in 3D space.
- **AMASS:** To initialize the spatio-temporal transformer backbone, we will use weights pre-trained on the AMASS dataset through Masked Pose Modeling. This dataset offers a diversity of 3D human motion sequences.
- **Fit3D:** This dataset is fitness-specific, containing 3D human poses with video sequences of exercises such as lunges, squats, etc. It allows use to have accurate 3D joint annotations during quantitative evaluation.

5 POTENTIAL RISKS

We must take care when mixing multiple datasets, as there may be discrepancies in annotated joint coordinates across datasets. Each dataset must be carefully analyzed to ensure some sort of standardized format when training Sáráandi et al. (2023), otherwise our model may fail to converge. We must also be careful when working with both 2D and 3D data during training, as miscalculating projection loss in residuals may undermine the model's accuracy.

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